



3D Printed Hyperloop

A 3D printing startup just took 24 hours to build a model of the Hyperloop
<http://dgit.in/FruitLoops>



Xbox launch titles

Launch titles list for the Xbox One has also been revealed. Take a look
<http://dgit.in/1549XCB>

From the labs

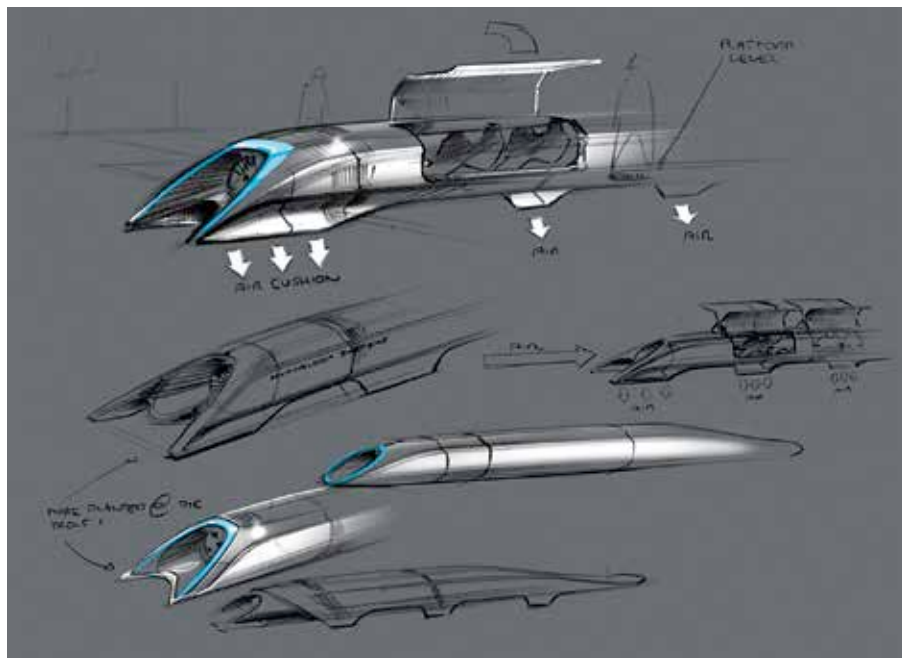
Travel times

The proposed route from Los Angeles to San Francisco is 960 kilometers long, a car takes five hours to cover that distance while current public transportation could take as much as 14 hours. Hyperloop claims to cover the same distance in approximately 35 minutes! For an Indian perspective into this scenario, a trip from Mumbai to Indore in just half an hour.

ance. There are a few ways of implementing the concept, one involves having the tube filled with air and the pods are pushed by the air pressure created behind each pod. This concept has been proven on a smaller scale with vacuum tubes which were used to transport small parcels and messages within a building, however, on a larger scale it's quite difficult to pull off and the friction due to the air present wouldn't have allowed for greater speeds than current generation bullet trains.

The other option is to reduce the volume of air within the tube to create a hard vacuum. This eliminates the problems of air compression and drag, however, maintaining vacuum in small contained areas is still a difficult prospect. Ensuring the same for a tube that spans the length of 350 miles is not possible given that there are going to be up to ten joints which will have to be sealed completely shut and even a simple crack anywhere could cause the section to implode. So with the Hyperloop, Elon Musk has proposed an environment of low pressure thus hitting a balance between the two extremes which were earlier proposed.

Each pod is to have an air compressor built onto it which takes air from the front of the pod and creates a compressed air cushion underneath the pod. This compensates for the Kármán limit which states that if the gap between the pod's outer walls and the tube's inner wall goes below a certain threshold then you'll have the equivalent of a giant syringe. The pod will have to push the air in front of it which in turn would use up more power. Then comes the trouble with



A schematic that's hand-drawn by Elon Musk. Nah we kid. But looks cool doesn't it?

“Speed has never killed anyone. Suddenly becoming stationary, that's what gets you.”
- Jeremy Clarkson

powering the air compressor using a battery on the pod and since these folks are the ones behind the Tesla electric car they handled the battery problem with ease. So to summarise, we have a pod with humans elevated by electromagnets, has a curtain of compressed air underneath it and has enough battery life to power the compressor the whole 350 miles (563 kilometers).

What could go wrong?

For starters, this is a massive straight tube across a vast landscape. In order to maintain a straight tube the gradient of the natural terrain must be compensated, this requires a lot of different shaped pillars / pylons. Mass manufacturing isn't exactly feasible given the varying heights of each pylon. Then comes into play thermodynamics which states that the increased pressure due to the air cushion would lead to an excess of heat to be generated thus making the tube an oven of sorts.

The pod is practically a bullet, if any of the pylons give away when a pod goes over it there is no easy way to stop the pod in time and save the lives of the people on board. Statistically, the casualties will be less given the low capacity per pod but even that wouldn't go down well with anyone.

Another surprising revelation was that Elon Musk was not interested in building the project on his own, he'd rather prefer it if several investors pitched in and made it a reality. Primarily because in the last two ventures that he began the initial concept and the final product were quite different, here you can't utilise the same approach given the enormous cost and waste of infrastructure that would arise by changing plans towards the end of the process.

Can he pull it off?

Not in the near future. There have been no prototypes yet to demonstrate the concept and a rapid transit system has already been approved along the proposed route. Mr. Musk has solicited suggestions on improving the current concept so we might see a few revisions of the idea before the execution phase gains momentum. Till then Mr. Musk will be taking the car to work. 